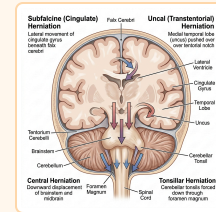


# Intracranial Hypertension & Herniation - Stroke

## CLINICAL SIGNS OF HERNIATION

- **Motor decline:** Spontaneous GCS motor score decrease of  $\geq 1$  point.
- **Pupillary reactivity:** Decrease in pupillary reactivity (Neurological Pupil Index, NPI  $< 3$ ).
- **Asymmetry:** New pupillary asymmetry or unilateral dilation (ipsilateral mydriasis).
- **Focal deficit:** New focal motor deficit or abnormal posturing (decorticate / decerebrate).
- **Cushing's Triad (Late Sign):** Systolic hypertension, bradycardia, and irregular respirations.  
**\*Cushing triad is a LATE sign of brainstem compression.\***



| Syndrome                      | Anatomical Substrate                                    | Clinical Exam & Diagnostic Trap                                                                                                                             |
|-------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Uncal</b> (Lateral)        | Medial temporal lobe (uncus) pushed over tentorial edge | Ipsilateral sluggish/dilated pupil (CN III compressed), contralateral hemiparesis. <b>Kernohan's Notch</b> causes false-localizing ipsilateral hemiparesis. |
| <b>Central</b> (Axial)        | Downward diencephalic and midbrain displacement         | Progressive stupor, midpoint fixed pupils, decorticate to decerebrate posturing. Symmetrical signs often confused with metabolic encephalopathy.            |
| <b>Subfalcine</b> (Cingulate) | Cingulate gyrus displaced under the falx cerebri        | Often clinically silent, or presents with contralateral lower extremity weakness. <b>ACA compression</b> causes frontal/leg territory infarction.           |
| <b>Tonsillar</b> (Downward)   | Cerebellar tonsils forced through the foramen magnum    | Cushing's triad, flaccid quadriplegia, respiratory arrest.                                                                                                  |

## MANAGEMENT

### GENERAL APPROACH

#### Fundamental Measures

- Elevate HOB 30°; strict neutral midline neck alignment to preserve venous outflow.
- Euvolemia (isotonic saline; avoid hypotonic D5W).
- Temperature  $< 38.0^{\circ}\text{C}$ .
- Normocapnia ( $\text{pCO}_2$  35–45 mmHg).
- When ICP is monitored, CPP = MAP - ICP; many protocols target CPP around  $>60$  mmHg, individualized to disease context.

#### Medical Interventions

- **Analgesia/sedation (fentanyl/propofol):** Target RASS -1 to +1 to prevent coughing, agitation, or ventilator dyssynchrony.
- **Mannitol 20%:** 1 g/kg IV bolus over 20–30 min. Must use in-line filter. **Hold if Osm  $> 320$  or Gap  $\geq 20$ .**
- **Hypertonic Saline (HTS):** 3% (150–250 mL bolus) or 23.4% (30 mL rescue bolus; central access only). **Hold if Na  $> 155$  or Cl  $> 115$ .**
- **Ventilation:** Maintain normocapnia ( $\text{PaCO}_2$  35–45 mmHg). For impending herniation only, use brief controlled hyperventilation targeting about 30–35 mmHg while definitive therapy is initiated; avoid prophylactic or prolonged hypocapnia.
- **Refractory ICP Elevation:** High-dose barbiturate therapy (pentobarbital) titrated to burst suppression on EEG.

### CSF DIVERSION

EVD placement for acute hydrocephalus, intraventricular hemorrhage (IVH), or mass effect with ventriculomegaly.

### DECOMPRESSIVE SURGERY

**Malignant MCA (DHC):** Age  $\leq 60$  years, clinical decline, infarct  $\geq 50\%$  MCA territory, within 48h (DECIMAL/DESTINY).

**Cerebellar Stroke:** Suboccipital craniectomy for brainstem compression, 4th ventricle effacement, or hydrocephalus.

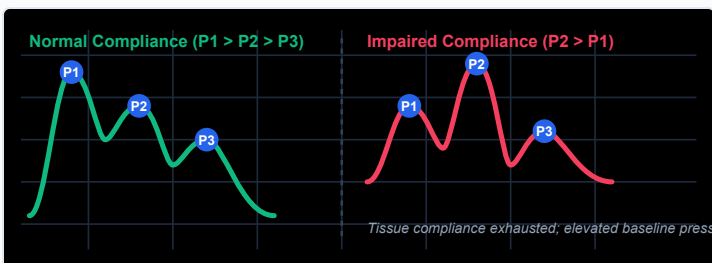
**Intracranial Hemorrhage (ICH):** Cerebellar ICH with deterioration, brainstem compression, hydrocephalus, or large size requires urgent surgical evaluation. Supratentorial/lobar ICH evacuation or decompression is case-dependent rather than routine.

### Management is not necessarily sequential

For active herniation or rapid clinical/radiographic deterioration, immediately initiate medical interventions & call Neurosurgery.

**Corticosteroids are not indicated for cytotoxic edema in stroke and increase infection risk.**

## ICP WAVEFORM ANALYSIS



- **P1 (Percussion wave):** Arterial pulsation.
- **P2 (Tidal wave):** State of intracranial compliance (elastic reserve).
- **P3 (Dicrotic wave):** Venous pulsations.
- **Compliance States:**
  - **Normal Compliance:**  $P1 > P2 > P3$  (elastic brain tissue easily cushions pulsations).
  - **Impaired Compliance / High ICP:**  $P2 > P1$  (brain tissue reserve exhausted; high risk of herniation).

### CPP = MAP - ICP

In patients with intracranial hypertension or mass effect, cerebral perfusion is highly pressure dependent, and cautious BP lowering is recommended.